Chapter 2

Inverse Laplace transforms:

If the Laplace transform of a function $f(t)$ is $F(s)$ i.e., if $\mathcal{L}\{f(t)\} = F(s)$ then $f(t)$ is called the inverse Laplace transforms of $F(s)$ and we write

$$\mathcal{L}^{-1}\{F(s)\} = f(t).$$

Important formulae of Inverse Laplace transformation:

1. $\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} = 1$
2. $\mathcal{L}^{-1}\left\{\frac{1}{s^{n+1}}\right\} = \frac{t^n}{n!}, n = 0,1,2, \ldots$
3. $\mathcal{L}^{-1}\left\{\frac{1}{s-a}\right\} = e^{at}$
4. $\mathcal{L}^{-1}\left\{\frac{s}{s^2-a^2}\right\} = \cosh at$
5. $\mathcal{L}^{-1}\left\{\frac{a}{s^2-a^2}\right\} = \sinh at$
6. $\mathcal{L}^{-1}\left\{\frac{s}{s^2+a^2}\right\} = \cos at$
7. $\mathcal{L}^{-1}\left\{\frac{a}{s^2+a^2}\right\} = \sin at$

Some workout examples on Inverse Laplace transformation:

Example 2.1

$$\mathcal{L}^{-1}\left\{\frac{s^2+1}{s^3}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s} + \frac{1}{s^3}\right\} = 1 + \frac{t^2}{2!} = 1 + \frac{t^2}{2}.$$

Example 2.2

$$\mathcal{L}^{-1}\left\{\frac{1}{2s-5}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{2(s-\frac{5}{2})}\right\} = \frac{1}{2}e^{\frac{5}{2}t}$$

Example 2.3

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2-16}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{4}\frac{4}{s^2-4^2}\right\} = \frac{1}{4}\sinh 4t$$

Example 2.4

$$\mathcal{L}^{-1}\left\{\frac{2s}{s^2-9}\right\} = 2\mathcal{L}^{-1}\left\{\frac{s}{s^2-3^2}\right\} = 2\cosh 3t$$

Example 2.5

$$\mathcal{L}^{-1}\left\{\frac{4}{s-2} - \frac{s}{s^2-16} + \frac{4}{s^2-4}\right\}$$
$$= 4\mathcal{L}^{-1}\left\{\frac{1}{s-2}\right\} - \mathcal{L}^{-1}\left\{\frac{s}{s^2-4^2}\right\} + 2\mathcal{L}^{-1}\left\{\frac{2}{s^2-2^2}\right\}$$
$$= 4e^{2t} - \cosh 4t + 2\sinh 2t.$$

Example 2.6

$$\mathcal{L}^{-1}\left\{\frac{5}{s} - \frac{3s}{s^2+16} + \frac{2}{s^2+4}\right\}$$
$$= 5\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} - 3\mathcal{L}^{-1}\left\{\frac{s}{s^2+16}\right\} + \mathcal{L}^{-1}\left\{\frac{2}{s^2+2^2}\right\}$$
$$= 5 - 3\cos 4t + \sin 2t.$$

First translation property:

If $\mathcal{L}^{-1}\{F(s)\} = f(t)$ then $\mathcal{L}^{-1}\{F(s-a)\} = e^{at}f(t)$.

Example 2.7

$$\mathcal{L}^{-1}\left\{\frac{10}{(s+3)^4}\right\}$$
$$= 10\mathcal{L}^{-1}\left\{\frac{1}{(s+3)^4}\right\}$$
$$= 10e^{-3t}\mathcal{L}^{-1}\left\{\frac{1}{s^4}\right\} = 10e^{-3t}\frac{t^3}{3!}.$$

Example 2.8

$$\mathcal{L}^{-1}\left\{\frac{1}{(s-2)^2+1}\right\}$$
$$= e^{2t}\mathcal{L}^{-1}\left\{\frac{1}{s^2+1}\right\}$$
$$= e^{2t}\sin t.$$

Example 2.9

$$\mathcal{L}^{-1}\left\{\frac{s-1}{(s-1)^2+4}\right\}$$
$$= e^{t}\mathcal{L}^{-1}\left\{\frac{s}{s^2+4}\right\}$$
$$= e^{t}\cos 2t.$$

Example 2.10

$$\mathcal{L}^{-1}\left\{\frac{s+2}{(s+2)^2-25}\right\}$$
$$= e^{-2t}\mathcal{L}^{-1}\left\{\frac{s}{s^2-25}\right\}$$
$$= e^{-2t}\cosh 5t.$$

Example 2.11

$$\mathcal{L}^{-1}\left\{\frac{1}{(s+3)^2-4}\right\}$$
$$= e^{-3t}\mathcal{L}^{-1}\left\{\frac{1}{s^2-2^2}\right\}$$
$$= e^{-3t}\mathcal{L}^{-1}\left\{\frac{1}{2}\frac{2}{s^2-2^2}\right\}$$
$$= \frac{1}{2}e^{-3t}\sinh 2t.$$

Example 2.12

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2+4s+13}\right\} = \mathcal{L}^{-1}\left\{\frac{s+2-2}{(s+2)^2+3^2}\right\}$$
$$= \mathcal{L}^{-1}\left\{\frac{s+2}{(s+2)^2+3^2}\right\} - \mathcal{L}^{-1}\left\{\frac{2}{(s+2)^2+3^2}\right\}$$
$$= e^{-2t}\,\mathcal{L}^{-1}\left\{\frac{s}{s^2+3^2}\right\} - 2\,e^{-2t}\mathcal{L}^{-1}\left\{\frac{1}{s^2+3^2}\right\}$$
$$= e^{-2t}\,\mathcal{L}^{-1}\left\{\frac{s}{s^2+3^2}\right\} - \frac{2}{3}e^{-2t}\mathcal{L}^{-1}\left\{\frac{3}{s^2+3^2}\right\}$$
$$= e^{-2t}\cos 3t - \frac{2}{3}e^{-2t}\sin 3t.$$

Example 2.13

$$\mathcal{L}^{-1}\left\{\frac{2s+1}{s^2+4s+13}\right\} = \mathcal{L}^{-1}\left\{\frac{2(s+2)-3}{(s+2)^2+9}\right\} = \mathcal{L}^{-1}\left\{\frac{2(s+2)-3}{(s+2)^2+9}\right\}$$
$$= \mathcal{L}^{-1}\left\{\frac{2(s+2)}{(s+2)^2+3^2} - \frac{3}{(s+2)^2+3^2}\right\} = 2e^{-2t}\cos 3t - 3\,e^{-2t}\sin 3t.$$

Example 2.14

$$\mathcal{L}^{-1}\left\{\frac{s}{(s+3)^5} - \frac{2s+7}{s^2+4s+29}\right\}$$
$$= \mathcal{L}^{-1}\left\{\frac{s+3-3}{(s+3)^5} - \frac{2(s+2)+3}{(s+2)^2+5^2}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{s+3}{(s+3)^5} - \frac{3}{(s+3)^5} - \frac{2(s+2)}{(s+2)^2+5^2} - \frac{3}{(s+2)^2+5^2}\right\}$$
$$= \mathcal{L}^{-1}\left\{\frac{1}{(s+3)^4} - \frac{3}{(s+3)^5} - 2\frac{(s+2)}{(s+2)^2+5^2} - \frac{3}{5}\frac{5}{(s+2)^2+5^2}\right\}$$
$$= e^{-3t}\frac{t^3}{3!} - 3e^{-3t}\frac{t^4}{4!} - 2e^{-2t}\cos 5t - \frac{3}{5}e^{-2t}\sin 5t\,.$$

Inverse Laplace transformation using partial fraction:

Example 2.15

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2-5s+6}\right\}$$
$$= \mathcal{L}^{-1}\left\{\frac{1}{(s-3)(s-2)}\right\}$$
$$= \mathcal{L}^{-1}\left\{\frac{1}{(s-3)} - \frac{1}{(s-2)}\right\}$$
$$= e^{3t} - e^{2t}.$$

Let, $\frac{1}{(s-3)(s-2)} \equiv \frac{A}{s-3} + \frac{B}{s-2}$

$\Rightarrow 1 = A(s-2) + B(s-3)$

$If\ s = 2, B = -1$ and $if\ s = 3, A = 1$

Example 2.16

$$\mathcal{L}^{-1}\left\{\frac{3s+1}{(s+1)(s^2+1)}\right\}$$
$$= \mathcal{L}^{-1}\left\{\frac{3s+1}{(s+1)(s^2+1)}\right\}$$
$$= \mathcal{L}^{-1}\left\{\frac{-1}{s+1} + \frac{s+2}{s^2+1}\right\}$$
$$= \mathcal{L}^{-1}\left\{\frac{-1}{s+1} + \frac{s}{s^2+1} + \frac{2}{s^2+1}\right\}$$
$$= -e^{-t} + \cos t + 2\sin t.$$

Let, $\frac{3s+1}{(s+1)(s^2+1)} \equiv \frac{A}{s+1} + \frac{Bs+C}{s^2+1}$
$\Rightarrow 3s + 1 = A(s^2+1) + (Bs+C)(s+1)$
Comparing both sides, we get
$A + B = 0, B + C = 3\ and\ A + C = 1$
By solving, we get
$A = -1, B = 1$ and $C = 2$

Example 2.17

$$\mathcal{L}^{-1}\left\{\frac{4s+5}{(s-1)^2(s+2)}\right\}$$
$$= \mathcal{L}^{-1}\left\{\frac{1/3}{s-1} + \frac{3}{(s-1)^2} - \frac{1/3}{s+2}\right\}$$
$$= \frac{1}{3}e^t + 3t\,e^t - \frac{1}{3}e^{-2t}.$$

Let, $\frac{4s+5}{(s-1)^2(s+2)} \equiv \frac{A}{s-1} + \frac{B}{(s-1)^2} + \frac{C}{s+2}$
$\Rightarrow 4s + 5 = A(s-1)(s+2) + B(s+2) + C(s-1)^2$
Comparing both sides, we get
$A + C = 0, A + B - 2C = 4$
$and - 2A + 2B + C = 5$
By solving, we get
$A = \frac{1}{3}, B = 3$ and $C = -\frac{1}{3}$.

Example 2.18

$$\mathcal{L}^{-1}\left\{\frac{3s^2+13s+26}{s(s^2+4s+13)}\right\}$$
$$= \mathcal{L}^{-1}\left\{\frac{2}{s} + \frac{s+5}{(s+2)^2+3^2}\right\}$$
$$= \mathcal{L}^{-1}\left\{\frac{2}{s} + \frac{s+2}{(s+2)^2+3^2} + \frac{3}{(s+2)^2+3^2}\right\}$$

Let, $\frac{3s^2+13s+26}{s(s^2+4s+13)} \equiv \frac{A}{s} + \frac{Bs+C}{s^2+4s+13}$
$\Rightarrow 3s^2 + 13s + 26 = A(s^2+4s+13) + (Bs+C)s$
Comparing both sides, we get
$A + B = 3, 4A + C = 13\ and\ 13A = 26$
By solving, we get
$A = 2, B = 1$ and $C = 5$.

$$= 2 + e^{-2t}\mathcal{L}^{-1}\left\{\frac{s}{s^2+3^2}\right\} + e^{-2t}\mathcal{L}^{-1}\left\{\frac{3}{s^2+3^2}\right\}$$

$$= 2 + e^{-2t}\cos 3t + e^{-2t}\sin 3t$$

Problem set: 2.1

1. Find the inverse Laplace transform of the following functions $(1-19)$:

Using direct formula

a. $F(s) = \frac{1}{s-5}$, **Ans:** $f(t) = e^{5t}$.

b. $F(s) = \frac{1}{s^5}$, **Ans:** $f(t) = \frac{t^4}{24}$.

c. $F(s) = \frac{s^3-5s^2+6}{s^4}$, **Ans:** $f(t) = t^3 - 5t + 1$.

d. $F(s) = \frac{2+4s}{s^2+25}$, **Ans:** $f(t) = 4\cos 5t + \frac{2}{5}\sin 5t$.

e. $F(s) = \frac{3}{s^2+4}$, **Ans:** $f(t) = \frac{3}{2}\sin 2t$.

f. $F(s) = \frac{3}{s^2-4}$, **Ans:** $f(t) = \frac{3}{2}\sinh(2t)$.

First Translation Property

g. $F(s) = \frac{1}{(s-3)^4}$, **Ans:** $f(t) = e^{3t}\ \frac{t^3}{6}$.

h. $F(s) = \frac{3}{(s+2)^2+9}$, **Ans:** $f(t) = e^{-2t}\sin 3t$.

i. $F(s) = \frac{s-2}{(s-2)^2-16}$, **Ans:** $f(t) = \frac{e^{-2t}}{2} + \frac{e^{6t}}{2}$. (Using $\cosh x = \frac{e^x + e^{-x}}{2}$.)

j. $F(s) = \frac{s}{s^2+4s-9}$, **Ans:** $f(t) = e^{-2t}\left(\cosh(\sqrt{13}\ t) - \frac{2\sqrt{13}\sinh(\sqrt{13}\,t)}{13}\right)$.

k. $F(s) = \frac{5s-7}{s^2-6s+25}$, **Ans:** $f(t) = 5\,e^{3t}\left(\cos 4t + \frac{2}{5}\sin 4t\right)$.

l. $F(s) = \frac{s}{s^2-6s+10}$, **Ans:** $f(t) = e^{3t}(\cos t + 3\sin t)$.

Using partial fraction

Type: unrepeated factors –

m. $F(s) = \frac{s+1}{s\,(s-2)\,(s+3)}$, **Ans:** $f(t) = \frac{3\,e^{2t}}{10} - \frac{2\,e^{-3t}}{15} - \frac{1}{6}$.

n. $F(s) = \frac{6}{(s+2)\,(s-4)}$, **Ans:** $f(t) = e^{4t} - e^{-2t}$.

o. $F(s) = \frac{6\,s-17}{s^2-5s+6}$, **Ans:** $f(t) = 5\,e^{2t} + e^{3t}$.

Type: repeated factors –

p. $F(s) = \frac{s}{(s+1)^2}$, **Ans:** $f(t) = e^{-t} - t\,e^{-t}$.

q. $F(s) = \frac{7\,s^2+14\,s-9}{(s-1)^2\,(s-2)}$, **Ans:** $f(t) = -40\,e^t - 12\,t\,e^t + 47\,e^{2t}$.

r. $F(s) = \frac{20}{(s^2+4s+1)\,(s+1)}$, **Ans:** $f(t) = 10\,e^{-2t}\left(\cosh\left(\sqrt{3}\,t\right) + \frac{\sqrt{3}\,\sinh\left(\sqrt{3}\,t\right)}{3}\right) - 10\,e^{-t}$.

s. $F(s) = \frac{s}{(s^2+4)\,(s-1)}$, **Ans:** $f(t) = \frac{2}{5}\sin 2t - \frac{1}{5}\cos 2t + \frac{1}{5}\,e^t$.

2. Problems from Book: Find the inverse Laplace transformation of the following functions assuming a, b, L, n as constants: (Advanced Engineering Mathematics – Erwin Kreyszig**)**

a. $\frac{s+2}{s^2+4}$
Ans: $\cos(2t) + \sin(2t)$

b. $\frac{3s+4}{s^2+4s+5}$
Ans: $3e^{-2t}\cos t - 2e^{-2t}\sin t$

c. $\frac{5\,s+1}{s^2-25}$
Ans: $5\cosh(5t) + \frac{1}{5}\sinh(5t)$

d. $\frac{3s}{s^2-2s+2}$
Ans: $3e^t\cos t$

e. $\frac{s+1}{s^4+9s^2}$
Ans: $\frac{1}{27}(3t - \sin(3t) - 3\cos(3t) + 3)$

f. $\frac{\omega\cos\theta + s\sin\theta}{s^2+\omega^2}$
Ans: $\sin(\omega t + \theta)$

g. $\frac{1}{(s+\sqrt{2})(s-\sqrt{3})}$
Ans: $\frac{-1}{\sqrt{3}+\sqrt{2}}e^{-\sqrt{2}t} + \frac{1}{\sqrt{3}+\sqrt{2}}e^{\sqrt{3}t}$

h. $\frac{6(s+1)}{s^4}$
Ans: $6\left(\frac{t^2}{2!} + \frac{t^3}{3!}\right)$

i. $\frac{12}{s^4} - \frac{228}{s^6}$
Ans: $12\frac{t^3}{3!} - 228\frac{t^5}{5!}$

j. $\frac{\pi}{(s+\pi)^2}$
Ans: $\pi t e^{-\pi t}$

k. $\frac{4s+32}{s^2-16}$
Ans: $4\cosh(4t) + 8\sinh(4t)$

l. $\frac{6}{(s+1)^3}$
Ans: $3e^{-t}t^2$

m. $\frac{s+10}{s^2-s-2}$

n. $\frac{21}{(s+\sqrt{2})^4}$

Ans: $e^{\frac{t}{2}}\cosh\left(\frac{3}{2}t\right)+7e^{\frac{t}{2}}\sinh\left(\frac{3}{2}t\right)$

Ans: $21e^{-\sqrt{2}t}\frac{t^3}{3!}$

o. $\frac{s+1}{s^2\,(s-3)}$

Ans: $\frac{1}{9}(-3t+4e^{3t}-4)$

p. $\frac{4}{s^2-2s-3}$

Ans: $2e^t\sinh(2t)$

q. $\frac{7.5}{s^2-2s-8}$

Ans: $2.5e^t\sinh(3t)$

r. $\frac{\pi}{s^2+10\pi s+24\pi^2}$

Ans: $e^{-5\pi t}\sinh(\pi t)$

s. $\frac{1/16}{s^2+s+\frac{1}{2}}$

Ans: $\frac{1}{8}e^{\frac{-1}{2}t}\sin\left(\frac{1}{2}t\right)$

t. $\frac{2s-1}{s^2-6s+18}$

Ans: $2e^{3t}\cos(3t)+\frac{5}{3}e^{3t}\sin(3t)$

u. $\frac{s-6.25}{s^2-6.25}$

Ans: $1.75e^{-2.5t}-0.75e^{2.5t}$

v. $\frac{3}{s^2+s/4}$

Ans: $3\left(4-4e^{\frac{-t}{4}}\right)$

w. $\frac{20}{s^3-2\pi s^2}$

Ans: $\frac{5(-2\pi t+e^{2\pi t}-1)}{\pi^2}$

x. $\frac{1}{s^3+as^2}$

Ans: $\frac{at+e^{-at}-1}{a^2}$

y. $\frac{1}{s^4-s^2}$

Ans: $\frac{1}{2}(-2t-e^{-t}+e^{t})$

z. $\frac{2s+1}{(s^2+1)\,(s-2)}$

Ans: $e^{2t}-\cos t$

Inverse Laplace transformation
Unit Step Function and Dirac's Delta Function:
Laplace transform of **unit step function** is

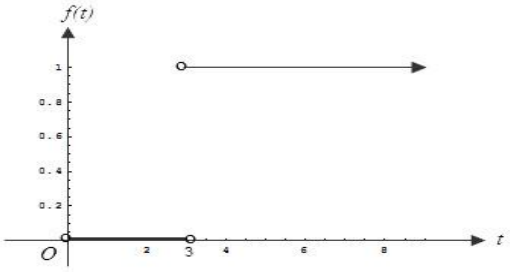
$$\mathcal{L}\{u(t-a)\} = \mathcal{L}\{u_a(t)\} = \frac{e^{-as}}{s}$$
$$\mathcal{L}\{f(t).u(t-a)\} = e^{-as}\mathcal{L}\{f(t+a)\}$$

So, $\mathcal{L}^{-1}\left\{\frac{e^{-as}}{s}\right\} = u(t-a) = u_a(t)$.
If $\mathcal{L}^{-1}\{F(s)\} = f(t)$, then $\mathcal{L}^{-1}\{e^{-as}F(s)\} = f(t-a)u_a(t) = f(t-a)u(t-a)$.

Some workout examples are given below:

Example 2.19

Find and sketch $f(t)$, where $f(t) = \mathcal{L}^{-1}\left\{\frac{e^{-3s}}{s}\right\}$.



Solution: we know that

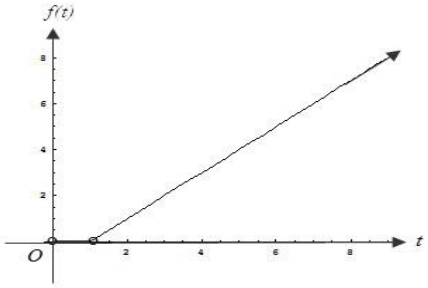
$$\mathcal{L}^{-1}\left\{\frac{e^{-as}}{s}\right\} = u(t-a) = u_a(t)$$

So,

$$f(t) = \mathcal{L}^{-1}\left\{\frac{e^{-3s}}{s}\right\}$$
$$= u(t-3) = u_3(t) = \begin{cases} 0, & t < 3 \\ 1, & t > 3 \end{cases}.$$

Example 2.20

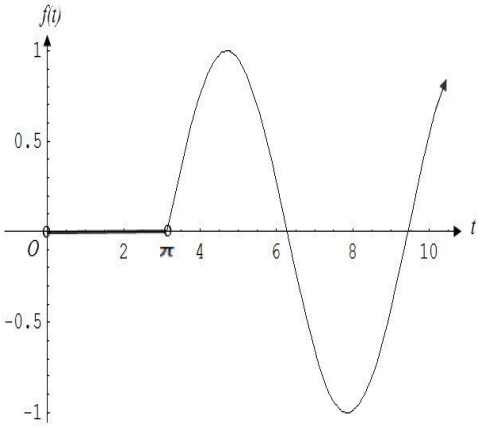
Find and sketch $f(t)$, where $f(t) = \mathcal{L}^{-1}\left\{\frac{e^{-s}}{s^2}\right\}$.



Solution:
Let, $F(s) = \frac{1}{s^2}$ and $\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} = t = f(t)$.
We know that,
$\mathcal{L}^{-1}\{e^{-as}F(s)\} = f(t-a)u_a(t)$
So, $\mathcal{L}^{-1}\left\{\frac{e^{-s}}{s^2}\right\} = f(t-1)u_1(t) = (t-1)u_1(t)$
$= \begin{cases} 0, & t < 1 \\ t-1, & t > 1 \end{cases}.$

Example 2.21

Find and sketch $f(t)$, where $f(t) = \mathcal{L}^{-1}\left\{\frac{e^{-\pi s}}{s^2+1}\right\}$.



Solution:
Let, $F(s) = \frac{1}{s^2+1}$ and $\mathcal{L}^{-1}\left\{\frac{1}{s^2+1}\right\} = \sin t = f(t)$.
We know that,
$\mathcal{L}^{-1}\{e^{-as}F(s)\} = f(t-a)u_a(t)$
So, $\mathcal{L}^{-1}\left\{\frac{e^{-\pi s}}{s^2+1}\right\} = f(t-\pi)u_\pi(t)$
$= \sin(t-\pi)\, u_\pi(t)$
$= \begin{cases} 0, & t < \pi \\ -\sin(\pi - t), & t > \pi \end{cases}$
$= \begin{cases} 0, & t < \pi \\ -\sin t, & t > \pi \end{cases}.$

Example 2.22

Find and sketch $f(t)$, where $f(t) = \mathcal{L}^{-1}\left\{\frac{e^{-s}}{s^2+\pi^2} + \frac{e^{-2s}}{s^2+\pi^2} + \frac{e^{-4s}}{s^2}\right\}$.

Solution:

$\mathcal{L}^{-1}\left\{\frac{1}{s^2+\pi^2}\right\} = \frac{1}{\pi}\sin(\pi t)$ and $\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} = t$.

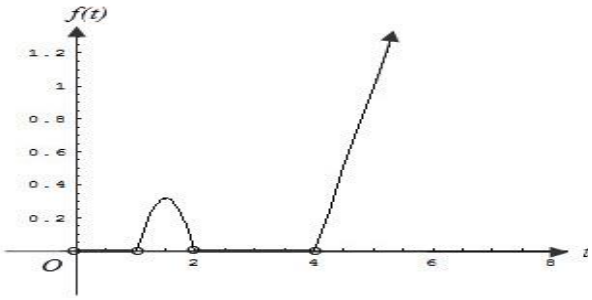
So, $f(t) = \mathcal{L}^{-1}\left\{\frac{e^{-s}}{s^2+\pi^2} + \frac{e^{-2s}}{s^2+\pi^2} + \frac{e^{-4s}}{s^2}\right\}$

$$= \frac{1}{\pi}\sin\big(\pi(t-1)\big)u_1(t) + \frac{1}{\pi}\sin\big(\pi(t-2)\big)u_2(t) + (t-4)u_4(t).$$

Since $\sin\big(\pi(t-1)\big) = -\sin(\pi t)$ and $\sin\big(\pi(t-2)\big) = \sin(\pi t)$, so the first two terms cancel each other when $t > 2$.

Hence, we obtain

$$f(t) = \begin{cases} 0, & 0 < t < 1 \\ -\frac{1}{\pi}\sin(\pi t), & 1 < t < 2 \\ 0, & 2 < t < 4 \\ t-4, & t > 4 \end{cases}.$$



Problem set 2.2

1. Find inverse Laplace of the following functions and sketch $f(t)$:

Associated with unit step function

a. $F(s) = 3\left(\frac{e^{-5s}}{s}\right)$ **Ans:** $f(t) = 3\,u_5(t) = \begin{cases} 0; & 0 < t < 5 \\ 3; & t > 5 \end{cases}$

b. $F(s) = 4\left(\frac{e^{-3s}}{s^2}\right)$ **Ans:** $f(t) = 4(t-3)\,u_3(t) = \begin{cases} 0 & ; 0 < t < 3 \\ 4\,(t-3); & t > 3 \end{cases}$

c. $F(s) = \frac{se^{-\pi s}}{s^2+25}$ **Ans:** $f(t) = -\cos(5t)\;u_\pi(t) = \begin{cases} 0; & 0 < t < \pi \\ -\cos 5t\,; & t > \pi \end{cases}$

d. $F(s) = \frac{2\left(e^{-3s}-3e^{-4s}\right)}{s}$ **Ans:** $f(t) = 2u_3(t) - 6u_4(t) = \begin{cases} 0, & 0 < t < 3 \\ 2, & 3 < t < 4 \\ -4, & t > 4 \end{cases}$.

e. $F(s) = \frac{5\left(e^{-\pi s}+e^{-2\pi s}\right)}{s^2+25}$ **Ans:** $f(t) = (-\sin 5t)\;u_\pi(t) + (\sin 5t)\;u_{2\pi}(t)$

$$= \begin{cases} 0, & 0 < t < \pi \\ -\sin 5t\,, & \pi < t < 2\pi \\ 0, & t > 2\pi \end{cases}$$

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Associated with Dirac's delta function

f. $F(s) = 1$ **Ans:** $f(t) = \delta(t)$.

g. $F(s) = e^{-3s}$ **Ans:** $f(t) = \delta(t-3)$.

h. $F(s) = 25\,e^{-2s}$ **Ans:** $f(t) = 25\,\delta(t-2)$.

2. Problems from Book: Find the inverse Laplace transformation associated with the unit step function (Advanced Engineering Mathematics – Erwin Kreyszig**)**

a. $\frac{e^{-3s}}{(s-1)^3}$ b. $\frac{e^{-3s}}{s^4}$ c. $\left(\frac{s+1}{s^2}\right)e^{-s}$

d. $\frac{6(1-e^{-\pi s})}{s^2+9}$ e. $\frac{2\left(e^{-s}-e^{-3s}\right)}{s^2-4}$ f. $\left(\frac{2s-10}{s^3}\right)e^{-5s}$